A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Scientific methodologies & sampling design - Set 1

Mark scheme

AQA spec 3.7.1; Research methods; textbook pp. 390-393

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change **1** Random sampling might leave gaps / miss the pattern – systematic sampling avoids bias while covering the gradient **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **systematic sampling** → sampling at regular intervals • **transect** → a line along which samples are taken to show change with distance • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data – wider intervals (eg 4 m) may miss peaks and troughs **1** Take increasing numbers of samples at one site and calculate the running mean – the number needed is where the mean stops fluctuating / stabilises **1** Calculate standard deviation to assess scatter: more variable data need more samples **1** Test the method (extraction technique, timing) for reliability before the main study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
01.3	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Remove surface vegetation – pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates **up to 2** Collect worms that come to the surface for a standardised time – count / identify / weigh **up to 2 Limitations:** solution may not percolate impermeable soils or reach deep worms – worms may move sideways or downwards – some may die or not respond – slow-moving worms missed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 8 nights **1** The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards – extra samples do not change it significantly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size **1** Sampling at different conditions / times would make results not comparable **1** Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Accuracy:** how close a recorded result is to the real value **1 Precision:** the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value • **precision** → readings close together / fine resolution	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level **1** Standardised techniques ensure results collected at different times, places or by different people are comparable **1** Other variables are controlled or monitored so their effect can be assessed – allows the study to be repeated / checked (reliability) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **correlation** → a relationship in which the two variables change together • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 5 points below. **1** Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated **1** Same slope / gradient – simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time **1** Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it **1** Repeat each treatment several times and calculate means (and standard deviation) **1** Plot soil loss against cover – statistical test (Spearman's rank / t-test) for significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **Spearman** → a test for correlation between two sets of ranked data • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
03.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Rainfall intensity / raindrop size / height **1** Soil texture, compaction, organic matter, depth **1** Permeability of the material beneath **1** Gradient **1** Root binding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Permeability** → how easily water passes through / lets water through **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** All variables except the independent variable can be standardised in the laboratory – in the field uncontrollable variables (weather) must be measured instead **1** Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 40-50 traps **1** Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Widely scattered results (large standard deviation) mean a single sample may not be representative **1** More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Same type and size of pitfall trap, set for the same length of time **1** Traps emptied at the same time of day / same weather conditions in both habitats **1** Same number of traps and same method of positioning (random or systematic) in each habitat **1** Other variables that could affect results (eg rainfall, temperature) monitored or controlled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **pitfall trap** → a sunken container that catches ground invertebrates • **random** → positions chosen using random numbers so the choice is not biased	3
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally **1** Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends **1** The interval between samples should match the rate at which the factor changes, found from a preliminary study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** t-test (or Mann-Whitney U if not normally distributed) **1** Comparing two means of counted / measured data from two independent samples **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** The difference is statistically significant (less than 1% probability that it arose by chance) **1** The null hypothesis (no difference between fields) is rejected – the organic field has significantly more earthworms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** A significant difference does not show a causal relationship **1** Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **causal** → one variable actually causing the change in the other **Do not credit** • 'organic farming uses no pesticides' - a restricted list of pesticides is permitted	2

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Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Scientific methodology:** identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • **Sample location:** random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias – interval chosen so variation is detected without excess data • **Sample timing:** diurnal, seasonal, weather-related and long-term changes – sampling on different occasions – interval matched to rate of change • **Sample size and number:** larger and more samples for non-homogeneous variables – running mean and standard deviation show when the mean is reliable – preliminary study to decide • Standardisation and control of variables – accuracy, precision, reliability, representativeness – anomalies and repeats – calibration of equipment – appropriate techniques and risk management • **Statistical analysis:** choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • **Evaluation:** trade-offs between cost, time, disturbance and reliability – limitations of sub-sampling – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

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Question	Marking guidance	Marks
	• **valid** → actually measures what it is meant to measure • **null hypothesis** → a statement that there is no significant difference or correlation • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	
	Model answer - Level 3 (8 marks)	
	A valid investigation begins with a clear, testable hypothesis and a null hypothesis, informed by existing knowledge, so that the data collected can actually accept or reject it. The first design decision is sample location: random sampling (grid coordinates or numbered sites) or systematic sampling along transects avoids the bias that would come from choosing convenient sites, and a preliminary study shows the interval needed to detect variation without collecting unnecessary data.	
	Timing matters because many variables change diurnally, seasonally or with weather, so samples on different occasions are needed and the interval must match the rate of change. Sample size and number depend on how variable the population is: a running mean shows when enough samples have been taken and the standard deviation shows the scatter around the mean. Techniques must be standardised so results from different sites, times or researchers are comparable, other variables controlled or monitored, and equipment calibrated so that precise readings are also accurate; anomalies should be repeated rather than ignored.	
	Finally the analysis must be chosen to suit the data - Spearman's rank for correlation, chi-squared for frequencies, a t-test for two means or Mann-Whitney U for medians - so that significance ($p < 0.05$) can be assessed, remembering that a significant correlation does not prove causation. Every decision trades reliability against cost, time and disturbance; sub-samples can only estimate the whole, so the most valid study is the one whose sampling design makes its limitations explicit.	
	Weaker answer - Level 2 (5 marks)	
	When planning fieldwork you need a hypothesis. You should choose sites randomly so it is not biased, for example using random numbers. You need to take enough samples and repeat them so the results are reliable. The method should be the same each time. You should measure accurately with calibrated equipment. Then you do a statistical test such as a t-test to see if the results are significant. If p is less than 0.05 the result is significant. This makes the conclusions valid.	
	What makes the difference	
	• Level 2 is sound but generic: it says 'enough samples' where Level 3 explains how a running mean and standard deviation tell you how many.	
	• Level 3 covers timing and the choice between statistical tests; Level 2 mentions one test only.	
	• The evaluation is missing from Level 2 - there is no discussion of trade-offs or of correlation vs causation, so it cannot reach the top level however accurate it is.	

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	

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